In this example, Hash 2 has a higher difficulty level than Hash 1 because it requires more zeros at the beginning. This means that it is harder to find a block hash that meets the target set by the network when the difficulty level is higher.

Finding a valid block hash involves **a lot of computer work.**

By finding a valid block hash, a **miner demonstrates that they have done the work required** to add the new block to the blockchain and get paid in bitcoin for their effort. **Proof of Work (PoW)** is the method ***Bitcoin*** uses to validate transactions and add new blocks to the blockchain.

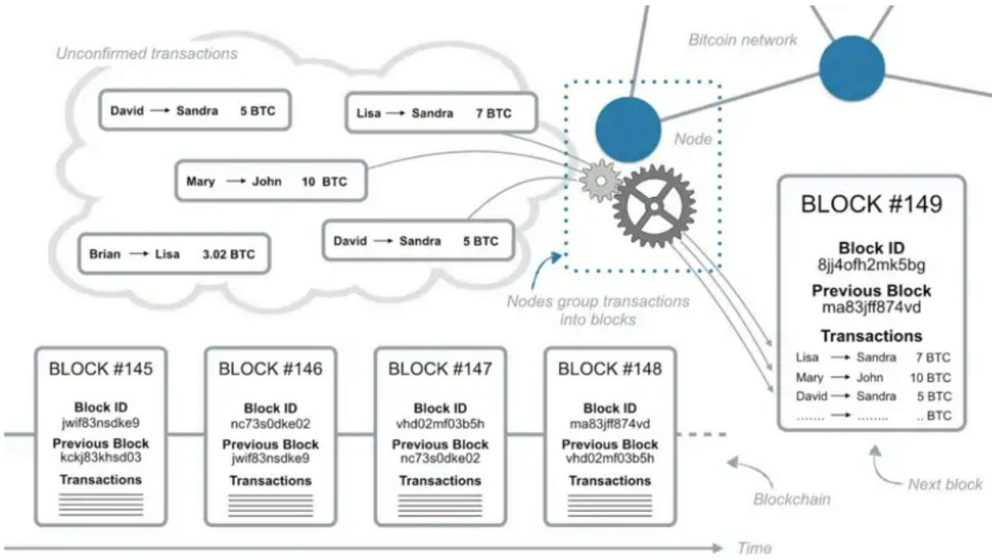
PoW keeps the *blockchain* safe by making it difficult for anyone with malicious intentions to take control. The target is adjusted to aim for one block to be mined every 10 minutes, making the network even safer as the target becomes harder to reach.

As an example, the target set by the network for mining a specific block could be:

00000000A1mINgF0RbL0cK5wItHth3hAy5tAcK

Which means that first miner to find a hash with eight leading zeros will achieve this target thus be allowed to add the block to the blockchain and will receive compensation in bitcoin.

The Role of Miners

Miners actually have two tasks in a *blockchain* network: 1) **verifying transactions** and 2) **adding new blocks**.

They collect unconfirmed transactions into their **mempool**, select a subset of these to include in their **candidate block**, and then search for the block hash.